

Probabilistic Deep Learning Frameworks for Forecasting Equity Returns

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Abstract

This paper investigates whether highly flexible deep learning models can accurately forecast daily equity-index returns. The primary forecasting target is the conditional quantile surface of cumulative SPY returns over one- and five-day horizons. To achieve this, three advanced residual models are evaluated: a self-supervised multi-resolution patch Transformer, a Tail-GAN optimized via a joint Value-at-Risk and Expected-Shortfall scoring rule, and a direct generative-objective model (GOM).

Crucially, rather than blindly substituting the statistical baseline, candidate forecasts are aggregated across three random seeds and filtered through a causal probabilistic regime gate. This gate is rigorously estimated on chronological validation blocks and requires models to demonstrate stable episode-level improvements, sufficient seed diversity, a positive lower bound in a circular-block-bootstrap test, and acceptable tail-risk calibration before activation.

During the development experiments, the regime gate successfully identified valuable non-linear patterns: GOM and Tail-GAN reduced the mean pinball loss relative to the EWMA reference by 0.408% and 0.285% in the volatile 2022 period, while Tail-GAN secured a smaller 0.203% reduction in 2024. No candidates met the promotion criteria during 2020. In the 2025 evaluation using 226 common forecast origins, the raw neural ensembles exhibited substantial ex post loss reductions compared to EWMA losses (0.252% for the Transformer, 1.790% for Tail-GAN, and 1.770% for GOM). Ultimately, these findings demonstrate that while neural residuals effectively capture complex market dynamics within selected regimes, establishing a universally stable, cross-regime selection rule remains a critical open challenge in quantitative finance.

Keywords: probabilistic forecasting; quantile regression; generative adversarial networks; self-supervised learning; regime switching; Value-at-Risk.

JEL classification: C22, C45, C53, G17.

1 Introduction

Forecasting financial returns is inherently difficult because weak predictability in average returns is frequently overshadowed by persistent volatility clustering, heavy-tailed distributions, and sudden, episodic dependencies across global markets. Consequently, relying on single point forecasts provides an incomplete and often misleading picture of true market uncertainty. A full conditional distribution, represented in this study by a comprehensive grid of quantiles, resolves this by supporting both central tendency forecasting and rigorous lower-tail risk measurement.

Flexible neural network architectures excel at capturing the nonlinear interactions that low-dimensional statistical recursions normally miss. However, this vast learning capacity introduces a significant risk of instability: a model that achieves exceptional performance during a specific market regime may completely fail to generalize, and this failure might not be detectable from the immediately preceding validation period. This crucial distinction forms the core philosophy of the present study. Here, neural networks are forbidden from replacing the statistical benchmark simply because they achieve a lower overall test loss. Instead, they are permitted to influence the final forecast only when a pre-determined promotion rule—calibrated entirely on chronological validation data—finds robust, statistically significant evidence of stability.

To operationalize this, the forecasting system is structured in two distinct layers. The first layer consists of a robust exponentially weighted Student- t distribution paired with an

adaptive bank of statistical models, forming a probabilistic baseline for comparison. The second layer deploys a patch Transformer, Tail-GAN, and GOM to estimate bounded, non-linear residual corrections relative to that baseline. A causal regime gate then decides whether these neural residuals are activated. If a neural model is rejected by the gate, the system seamlessly reproduces the statistical benchmark exactly.

The empirical design of this study is structured across three chronological development tests (2020, 2022, and 2024) and one verification test (2025). The development tests are used to diagnose the network architectures and refine the promotion rule. The verification test rigorously evaluates the system by retaining the exact same feature constructions, seed sets, quantile grids, model classes, and promotion thresholds. This design provides a direct assessment of both the raw forecast quality and the practical ability of the frameworks to successfully identify transferable model improvements. By adopting different parameters in different market scenarios, the models learn to optimize the performance.

Forecasting architecture and information flow

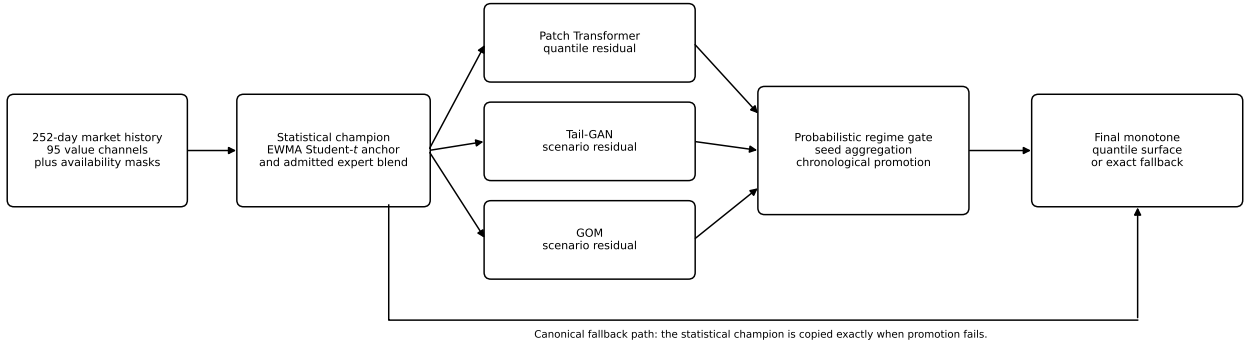


Figure 1: Comprehensive Forecasting Architecture. The statistical model acts as a continuous baseline comparison at every origin. Simultaneously, three distinct neural models estimate candidate residual corrections. The probabilistic regime gate acts as the final decision-maker, determining if the validation evidence justifies integrating those neural residuals into the final quantile surface, or if the system must fall back to the statistical baseline.

Main Contributions:

1. A adaptive forecasting framework where all candidate model families share an identical quantile reference, ensuring that rejected models revert to the statistical baseline.
2. The design of a targeted patch Transformer using multi-scale temporal tokenization, masked reconstruction, distribution pretraining, and a quantile-residual head (initialized to zero) to prevent erratic baseline deviations.
3. Implementation of constrained Tail-GAN and GOM scenario generators tied to VaR and ES scoring functions.
4. A strictly chronological promotion mechanism that decouples ex ante admissibility from ex post performance, proven via an uncorrupted 2025 evaluation.

2 Related Literature

2.1 Generative Forecasting and Tail-Risk Simulation

Generative adversarial networks (GANs) conceptualize complex distribution learning as a competitive minimax game between a generator and a discriminator (Goodfellow et al., 2014). Subsequent advancements, such as Wasserstein objectives, gradient penalties, and spectral normalization, greatly improved the geometric stability of this adversarial optimization (Arjovsky et al., 2017; Gulrajani et al., 2017; Miyato et al., 2018). Adapting this to time series, ForGAN demonstrated how conditional GANs could effectively handle probabilistic forecasting (Koochali et al., 2019), while Fin-GAN successfully combined adversarial learning with financial performance objectives, proving the necessity of integrating the generator, loss function, and evaluation pipeline (Vuletić et al., 2024).

Building on these foundations, Tail-GAN specifically targets the generation of portfolio tail-risk scenarios. Its primary advantage stems from the joint elicibility of Value-at-Risk (VaR) and Expected Shortfall (ES), which allows both the generator and the critic to optimize against a proper, financially coherent joint scoring rule (Fissler and Ziegel, 2016; Cont et al., 2025). The implementation studied in this paper retains this elicitable scoring principle but uniquely forces the generated scenarios to remain constrained around the EWMA model, thereby preventing the generation of economically impossible paths. GOM leverages this same differentiable

tail score but streamlines the architecture by removing the adversarial discriminator.

2.2 Transformers and Self-Supervised Time-Series Representations

The scaled attention mechanism revolutionized deep learning by allowing models to construct highly dynamic, data-dependent interactions among sequential tokens (Vaswani et al., 2017). In the context of time-series data, architectures like PatchTST revealed that grouping contiguous time-steps into distinct "patches" shortens the overall attention sequence length, drastically reducing computational complexity while beautifully preserving local temporal semantics (Nie et al., 2023). Furthermore, techniques like masked reconstruction and contrastive agreement provide robust, complementary signals for self-supervised representation learning (He et al., 2022; Chen et al., 2020). The specific Transformer model introduced here synergizes these concepts by merging three distinct patch lengths, specifically aligned with weekly, monthly, and quarterly market cycles.

2.3 Quantile Forecasts and Dependent Evaluation

Quantile regression relies on minimizing the asymmetric pinball loss function (Koenker and Bassett, 1978). Because independently predicted conditional quantiles can cross one another (creating an invalid probability distribution), monotone rearrangement is applied to restore a valid, monotonically increasing quantile function, a process proven to possess highly favorable approximation properties (Chernozhukov et al., 2010).

Evaluating multi-day financial forecasts introduces significant serial dependence because adjacent forecast origins inherently share market conditions and overlapping future returns. To conduct valid statistical inference, confidence intervals in this study are constructed using circular block resampling (Politis and Romano, 1992), closely related to the stationary bootstrap introduced by Politis and Romano (1994). Finally, the calibration of VaR estimates is rigorously assessed via the unconditional coverage test of Kupiec (1995) and the

exception-independence test of Christoffersen (1998).

3 Forecasting Problem and Data

3.1 Targets and the Scoring Rule

Let P_t represent the adjusted closing price of the SPY ETF at time t . The one-day logarithmic return is calculated as:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right). \quad (1)$$

For a designated forecasting horizon $h \in \mathcal{H} = \{1, 5\}$, the cumulative forecast target observed from origin t is the sum of the subsequent returns:

$$y_{t,h} = \sum_{j=1}^h r_{t+j}. \quad (2)$$

Rather than predicting a single mean expectation, this paper estimates the conditional quantile $Q_{t,h}(\tau)$. For any probability level τ , this represents the value q such that the probability of the actual return falling below q , given the available information $\mathcal{F}_t$, is exactly τ :

$$Q_{t,h}(\tau) = \inf\{q : \Pr(y_{t,h} \leq q \mid \mathcal{F}_t) \geq \tau\}. \quad (3)$$

To capture both the central tendency and the extreme tails, the model evaluates a dense probability grid:

$$Q = \{0.01, 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.975, 0.99\}.$$

To evaluate the accuracy of these conditional quantiles, this paper employs the asymmetric pinball loss function. For a given residual error $u = y - q$ (the difference between the realized target and the predicted quantile), the pinball loss at quantile level τ is defined as:

$$\rho_\tau(u) = u(\tau - \mathbb{I}_{\{u < 0\}}). \quad (4)$$

This function inherently penalizes over-predictions and under-predictions asymmetrically, effectively training the model to respect the target probability threshold τ . The aggregate reported score is the mean pinball loss averaged across all forecast origins, horizons, and quantile levels:

$$\mathcal{L}(Q) = \frac{1}{|\mathcal{T}||\mathcal{H}||Q|} \sum_{t \in \mathcal{T}} \sum_{h \in \mathcal{H}} \sum_{\tau \in Q} \rho_\tau(y_{t,h} - Q_{t,h}(\tau)). \quad (5)$$

To guarantee mathematical validity, the predicted quantiles are monotonically rearranged at each discrete coordinate (t, h) , strictly following Chernozhukov et al. (2010).

3.2 Predictor Panel Construction

Each individual forecast is conditioned on a trailing 252-trading-day history. This comprehensive, long-history panel is composed of 95 distinct value channels accompanied by a matching set of binary availability masks. These channels incorporate SPY price dynamics, cross-market ETF returns and volatilities, macroeconomic levels and differences, alongside lagged equity factor data. Macroeconomic indicators

include market volatility, Treasury yield curves, the federal funds rate, critical credit and financial-stress indices, oil prices, and the broad US dollar index. To avoid the look-ahead bias, every external macroeconomic series is shifted forward by a conservative release lag before being aligned to the standard trading calendar. Any missing observations are represented using availability masks.

3.3 Chronological Folds and Boundary Protection

The three primary development tests correspond to the years 2020, 2022, and 2024, with their respective validation blocks in 2019, 2021, and 2023. The conclusive final experiment uses data spanning through 2023 for model fitting, allocates 2024 exclusively for validation and gating, and tests on the unseen 2025 period.

Because the $y_{t,5}$ target encompasses five days of future returns, strict boundary protection must be enforced. A specific forecast origin is assigned to a data block only if its entire target window concludes within that block. Formally, for a data block terminating on date d , the set of valid origins is:

$$\mathcal{T}(d) = \{t : t + h_{\max} \leq d\}, \quad h_{\max} = 5. \quad (6)$$

Consequently, the final five candidate origins immediately preceding any split boundary are excluded, systematically preventing a training or validation target from containing forward-looking returns from a subsequent block.

4 Forecasting Models

4.1 EWMA Student- t Reference

The core statistical reference relies on a robust Exponentially Weighted Moving Average (EWMA) formulation. The conditional location (mean) μ_t and conditional variance σ_t^2 are updated recursively:

$$\mu_t = \lambda_\mu \mu_{t-1} + (1 - \lambda_\mu) r_t, \quad (7)$$

$$\sigma_t^2 = \lambda_\sigma \sigma_{t-1}^2 + (1 - \lambda_\sigma) (r_t - \mu_t)^2, \quad (8)$$

using decay parameters $\lambda_\mu = 0.985$ and $\lambda_\sigma = 0.94$. The variance decay matches the established standard daily RiskMetrics parameter (J.P. Morgan/Reuters, 1996). Letting $T_\nu^{-1}(\tau)$ represent the inverse cumulative distribution function of a Student- t distribution with ν degrees of freedom (estimated during training), the horizon- h reference quantile is defined as:

$$Q_{t,h}^A(\tau) = h\mu_t + \sqrt{h} \sigma_t T_\nu^{-1}(\tau). \quad (9)$$

This statistical recursion is highly advantageous: it severely limits estimation variance, rapidly adapts to shifting conditional volatility, and structurally accommodates heavier tail events than standard Gaussian assumptions allow.

4.2 The Adaptive Statistical Model

To strengthen the baseline, an adaptive pool aggregates multiple EWMA Student- t models, featuring diverse decay rates,

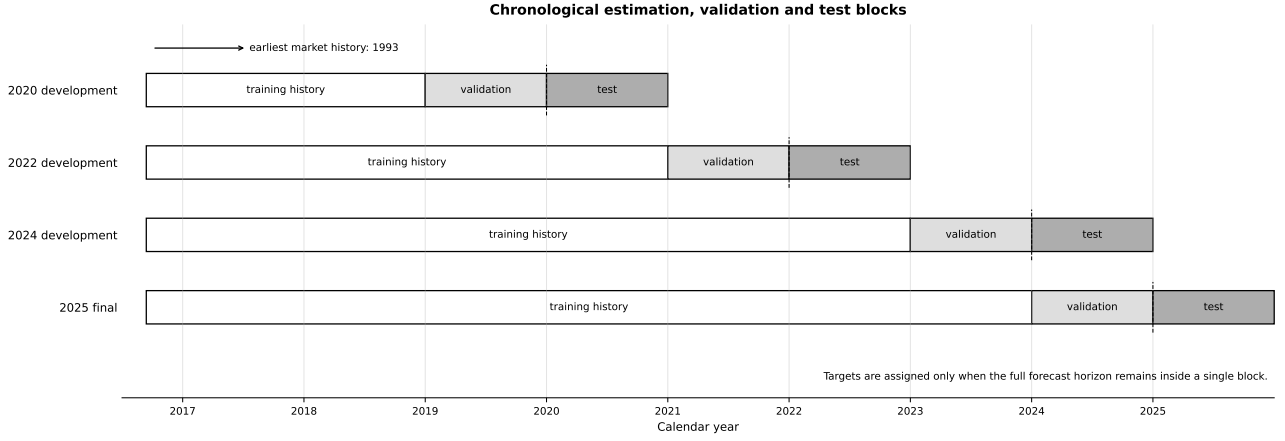


Figure 2: Strict Chronological Boundary Design. The diagram illustrates the segmentation of estimation, validation, and test blocks across recent calendar years. Boundary exclusions are rigorously enforced so that the 5-day overlapping targets never bridge the gap between training and testing environments.

leverage sensitivities, and asymmetric tail scales. The convex weights tailored to horizons blend these model estimates:

$$Q_{t,h}^P(\tau) = \sum_{j=1}^J a_{h,j} Q_{t,h}^{(j)}(\tau), \quad a_{h,j} \geq 0, \quad \sum_j a_{h,j} = 1. \quad (10)$$

The best statistical model is then formulated by blending this adaptive pool with the fixed reference baseline:

$$\tilde{Q}_{t,h}^S = m_{t,h} + s_h [Q_{t,h}^A + \beta_h (Q_{t,h}^P - Q_{t,h}^A) - m_{t,h}] + \delta_h, \quad (11)$$

where $m_{t,h} = Q_{t,h}^A(0.5)$ represents the median, $\beta_h \in [0, 1]$ controls the blending ratio, $s_h > 0$ scales the spread, and δ_h applies a localized, shrunk quantile offset. After ensuring monotonicity via the rearrangement operator $\mathcal{M}$, the final statistical model is determined as:

$$Q_{t,h}^S = \mathcal{M}(\tilde{Q}_{t,h}^S). \quad (12)$$

These internal weights, blends, and spread parameters are carefully calibrated across four disjoint, chronological training segments. Importantly, the adaptive components are allowed to influence the statistical model only when confirmation episodes and a one-sided block bootstrap provide rigorous statistical support.

4.3 Target-Relevant Patch Transformer

The neural Transformer aims to decode complex market contexts. Let $X_t \in \mathbb{R}^{L \times 2C}$ represent the standardized history of values and their masks, spanning $L = 252$ days and $C = 95$ channels. The localized temporal information is extracted by defining multi-scale patches. At a specific scale m , governed by a patch length $\ell_m \in \{5, 21, 63\}$ and a stride $s_m \in \{5, 10, 21\}$, the j -th patch vector is extracted via:

$$x_{t,j}^{(m)} = \text{vec}(X_t[j s_m : j s_m + \ell_m - 1, :]). \quad (13)$$

These raw values and their corresponding mask vectors are then linearly projected and fused with feature, scale, and positional embeddings to form a comprehensive token $e_{t,j}^{(m)}$:

$$e_{t,j}^{(m)} = W_v^{(m)} x_{t,j}^{(m)} + W_a^{(m)} a_{t,j}^{(m)} + u_m + p_j. \quad (14)$$

Following a mask-aware channel attention pooling phase, these tokens enter the Transformer’s self-attention mechanism. For the mapped query (Q), key (K), and value (V) matrices, each of the four attention heads computes dynamically scaled importance weights:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \quad (15)$$

The full architecture uses three encoder layers featuring a 128-dimensional hidden representation.

To force the model to learn deep structural patterns prior to supervised forecasting, the network undergoes self-supervised pretraining using two independently corrupted (masked) views of each history. The overarching pretraining objective combines four distinct loss components:

$$\mathcal{L}_{\text{SSL}} = \mathcal{L}_{\text{rec}} + \lambda_r \mathcal{L}_{\text{reg}} + \lambda_f \mathcal{L}_{\text{future}} + \lambda_c \mathcal{L}_{\text{con}}, \quad (16)$$

using weightings $(\lambda_r, \lambda_f, \lambda_c) = (0.20, 0.35, 0.10)$. The reconstruction ($\mathcal{L}_{\text{rec}}$) and future-distribution ($\mathcal{L}_{\text{future}}$) targets use a smooth- L_1 loss, while the regime descriptor ($\mathcal{L}_{\text{reg}}$) is penalized via standard squared error. The contrastive consistency loss ($\mathcal{L}_{\text{con}}$) aligns the representations of the two augmented views $z_t^{(1)}$ and $z_t^{(2)}$ using cosine similarity:

$$\mathcal{L}_{\text{con}} = 1 - \frac{z_t^{(1)\top} z_t^{(2)}}{\|z_t^{(1)}\|_2 \|z_t^{(2)}\|_2}. \quad (17)$$

Crucially, the future targets (cumulative, absolute, and down-side returns) act purely as supervisory signals during training; they are never supplied as encoder inputs, preserving causal validity.

When transitioning to supervised forecasting, the network’s output head calculates bounded parameters: location (ℓ), lower-scale (s_-), upper-scale (s_+), and curvature (p). By leveraging the established statistical model’s median ($m = Q^S(0.5)$), inter-quantile range ($R = Q^S(0.99) - Q^S(0.01)$), and centered deviations ($d_\tau = Q^S(\tau) - m$), the Transformer defines the new candidate quantile as:

$$Q^T(\tau) = m + \ell R + s_\tau \text{sgn}(d_\tau) R \left(\frac{|d_\tau|}{R}\right)^p, \quad (18)$$

$$s_\tau = \begin{cases} s_-, & \tau < 0.5, \\ s_+, & \tau \geq 0.5. \end{cases} \quad (19)$$

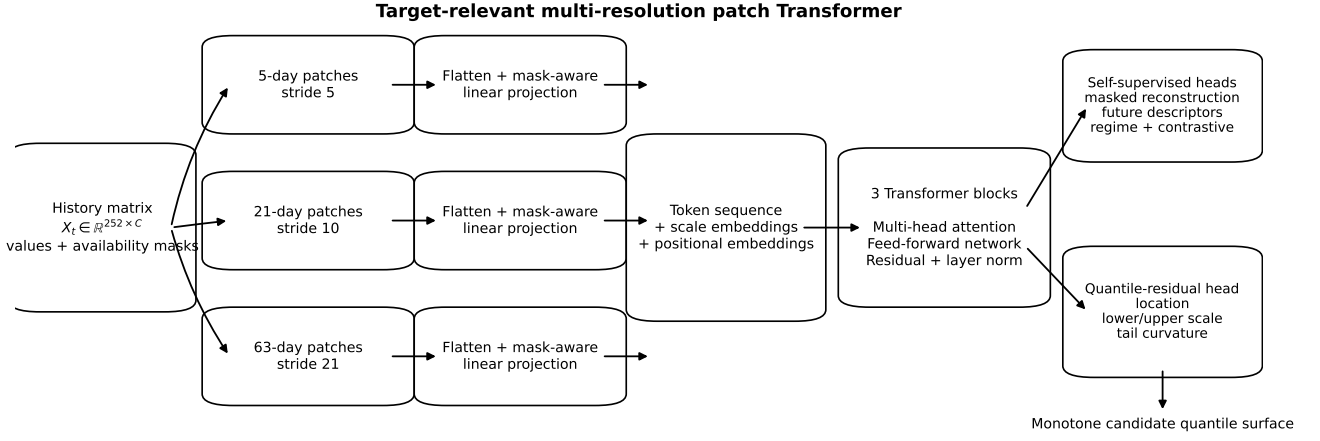


Figure 3: Multi-Resolution Patch Transformer Architecture. The model fragments historical data into weekly, monthly, and quarterly patches, subsequently projecting them into a unified token space. The Transformer encoder engages in complex self-supervised representation learning before a constrained quantile-residual head outputs the forecast corrections.

To enforce baseline comparison, this parameter head is strictly initialized at zero. Consequently, before any supervised weight updates occur, the initial candidate Q^T equals the statistical model Q^S . During supervised training, the network minimizes a tail-weighted pinball loss alongside parameter regularization, an auxiliary regime loss, and a critical benchmark guard penalty:

$$\lambda_g \max\{\mathcal{L}(Q^T) - \mathcal{L}(Q^S), 0\}. \quad (20)$$

This guard aggressively penalizes the Transformer if its modifications result in a loss worse than the simple statistical baseline.

4.4 Tail-GAN and GOM Generative Scenarios

Traditional Conditional GANs model data distributions by optimizing a minimax zero-sum game between a generator (G_θ) and a discriminator (D_ψ) conditional on context c :

$$\min_{\theta} \max_{\psi} \{ \mathbb{E}_{x \sim p_{\text{data}}} \log D_\psi(x | c) + \mathbb{E}_{z \sim p_z} \log[1 - D_\psi(G_\theta(z, c) | c)] \}. \quad (21)$$

However, the Tail-GAN implemented in this paper replaces standard binary classification with a deeply financial VaR-ES joint scoring rule. For a lower-tail probability α , true observation y , predicted VaR v , and predicted ES e , the elicitable score function is defined as:

$$S_\alpha(v, e; y) = \frac{w}{2} (\mathbb{I}_{\{y \leq v\}} - \alpha)(y^2 - v^2) + \mathbb{I}_{\{y \leq v\}} e(v - y) + \alpha e \left(\frac{e}{2} - v \right). \quad (22)$$

This score is systematically averaged over a predefined bank of static and dynamic trading strategies across crucial tail levels $\alpha \in \{0.01, 0.05, 0.10\}$.

To constrain the generator from producing implausible scenarios, it is not permitted to generate returns out of pure noise. Instead, it systematically modifies a covariance factor model:

$$\tilde{r}_{t,j} = \mu_t + d_t + B_t D_f(s_t) z_{t,j}^f + D_\varepsilon(u_t) z_{t,j}^\varepsilon + \kappa_t \min(z_{t,j}, 0). \quad (23)$$

In this structured perturbation, B_t is a rank-four loading matrix, z^f and z^ε are independently drawn Student- t shocks, and the terms s_t, u_t, d_t, κ_t are bounded origin-level parameters controlling scale, drift, and downside asymmetry, derived from a two-layer Gated Recurrent Unit (GRU).

Letting P_{real} and P_θ denote the vectors of observed and generated strategy P&Ls respectively, the Tail-GAN critic estimates accurate VaR-ES pairs. Its loss objective aims to minimize the discrepancy in these risk estimations:

$$\mathcal{L}_D = \lambda_D S(D_\psi(P_{\text{real}}); P_{\text{real}}) - S(D_\psi(P_\theta); P_{\text{real}}) + \eta_D \Omega(\psi), \quad (24)$$

where Ω alongside spectral normalization constrains the critic's magnitude and Lipschitz continuity (Miyato et al., 2018). The generator is subsequently trained to pass this critic by minimizing the overall risk discrepancy plus structural penalties:

$$\mathcal{L}_G^{\text{Tail}} = S(D_\psi(P_\theta); P_{\text{real}}) + \lambda_{\text{corr}} \mathcal{L}_{\text{corr}} + \lambda_{\text{acf}} \mathcal{L}_{\text{acf}} + \lambda_{\text{marg}} \mathcal{L}_{\text{marg}} + \lambda_{\text{en}} \mathcal{L}_{\text{en}}. \quad (25)$$

The Generative Objective Model (GOM) acts as a specialized ablation, completely removing the adversarial critic. Instead, differentiable soft quantiles and tail means are computed directly from the generated P&L vectors:

$$\mathcal{L}_G^{\text{GOM}} = S(\widetilde{\text{VaR}}_\alpha(P_\theta), \widetilde{\text{ES}}_\alpha(P_\theta); P_{\text{real}}) + \sum_k \lambda_k \mathcal{L}_k. \quad (26)$$

Both networks share sophisticated regularization penalties that compare generated and observed correlation matrices, autocorrelations, and marginal quantiles. Furthermore, an energy distance term $\mathcal{L}_{\text{en}}$ minimizes the statistical distance between distributions (Székely and Rizzo, 2013):

$$\mathcal{L}_{\text{en}} = 2\mathbb{E}\|X - Y\|_2 - \mathbb{E}\|X - X'\|_2 - \mathbb{E}\|Y - Y'\|_2. \quad (27)$$

4.5 The Causal Probabilistic Regime Gate

The regime gate benefits the model performance. Using strict thresholds, recent market features (e.g., volatility ratios, downside variation, drawdown vectors, VIX, credit spreads,

Constrained Tail-GAN and direct generative-objective model

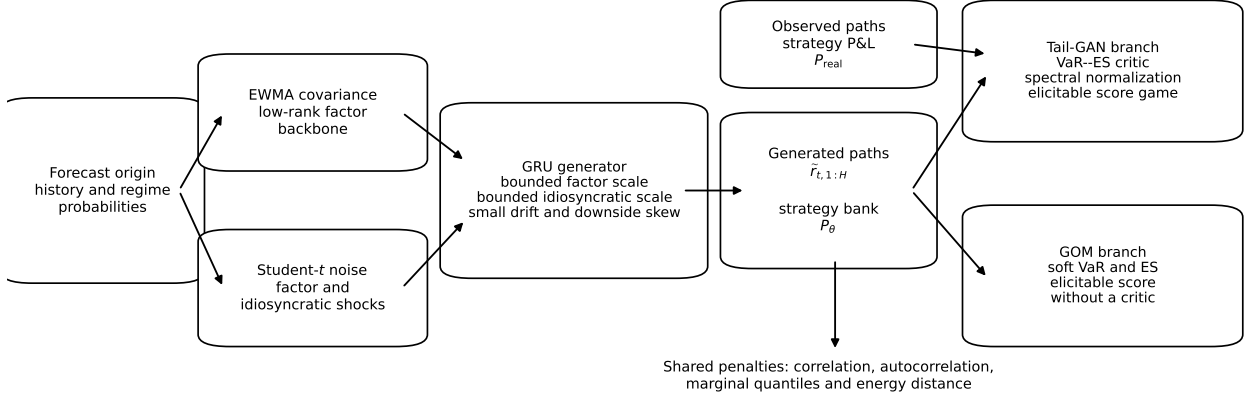


Figure 4: Constrained Scenario Generation Flow. Both generative branches (Tail-GAN and GOM) rely on a common recurrent factor-scale generator and share identical distributional penalties. The great divergence occurs in the evaluation of risk: Tail-GAN trains a dynamic critic to estimate VaR and ES, whereas GOM computes differentiable risk functionals directly from the generated paths.

and market dispersion) are transformed into a normalized vector of regime probabilities:

$$\mathbf{p}_t = (p_{t,c}, p_{t,m}, p_{t,s}), \quad p_{t,k} \geq 0, \quad \sum_k p_{t,k} = 1, \quad (28)$$

representing calm, transition, and stress states, respectively. To prevent erratic flipping, these probabilities are smoothed forward through time with a high persistence factor of 0.85. Let $c_t = \max_k p_{t,k}$ denote the maximum regime confidence, $c_0 = 0.45$ the absolute confidence floor, and w_k the validation-selected residual scaling caps. The model dynamically computes a final activation scalar α_t :

$$\alpha_t = \text{clip}\left(\frac{c_t - c_0}{1 - c_0}, 0, 1\right) \sum_k p_{t,k} w_k. \quad (29)$$

This activation gracefully controls the magnitude of the neural intervention. For a fully seed-weighted candidate output Q^M , the final, deployed forecast dynamically blends the statistical baseline Q^S with the neural model depending entirely on the activation state α_t :

$$Q_{t,h}^G(\tau) = \mathcal{M}[Q_{t,h}^S(\tau) + \alpha_t \{Q_{t,h}^M(\tau) - Q_{t,h}^S(\tau)\}]. \quad (30)$$

5 Estimation and Promotion Algorithms

The exact algorithmic steps ensuring strict causal integrity and robust statistical promotion are detailed in Algorithms 1, 2, and 3.

6 Empirical Results

6.1 Development Folds Analysis

Table 1 aggregates the benchmark, the raw ensemble capability, and the formal final score post-gating for every model across the development periods.

The most robust development success occurred during the highly volatile 2022 period. GOM successfully reduced the mean pinball loss from 0.0041258 down to 0.0041090, while Tail-GAN brought the loss down to 0.0041140. Both models were set on equal seed weights (yielding $N_{\text{eff}} = 3$), and critically, both passed the multi-episode validations and dependent-bootstrap strictures. Consequently, the 95% confidence intervals for average daily loss reduction remained positive for both accepted 2022 forecasts. While Tail-GAN also satisfied promotion criteria in 2024, its test interval straddled zero, indicating a less precise point estimate for the realized gain.

The 2020 test highlights the essential distinction between raw ex post test capability and formal ex ante model selection. Every raw neural ensemble generated in 2020 outperformed the EWMA baseline.

6.2 Untouched 2025 Evaluation

The final comparison spans 226 distinct out-of-sample dates evaluated across two horizons, resulting in 452 common horizon-origin observations per evaluated model. Table 2 clearly delineates raw 2025 test performance from the formal forecast rigorously determined by the 2024 promotion evidence.

The Tail-GAN and GOM substantially reduced the ex post raw test loss by approximately 1.8%, while the Transformer achieved a modest 0.25% raw gain.

6.3 Horizon and Tail-Risk Diagnostics

The substantial raw gains observed in 2025 were heavily concentrated at the extended five-day horizon. Specifically, Tail-GAN improved raw pinball loss by 1.00% at the one-day mark but surged to a 2.17% improvement at five days. GOM exhibited parallel behavior, improving by 1.20% and 2.04% respectively.

Table 4 analyzes the lower-tail risk diagnostics. The formal EWMA forecast yielded a 5% VaR exception rate of 7.52% for one-day horizons and 10.18% for five-day horizons.

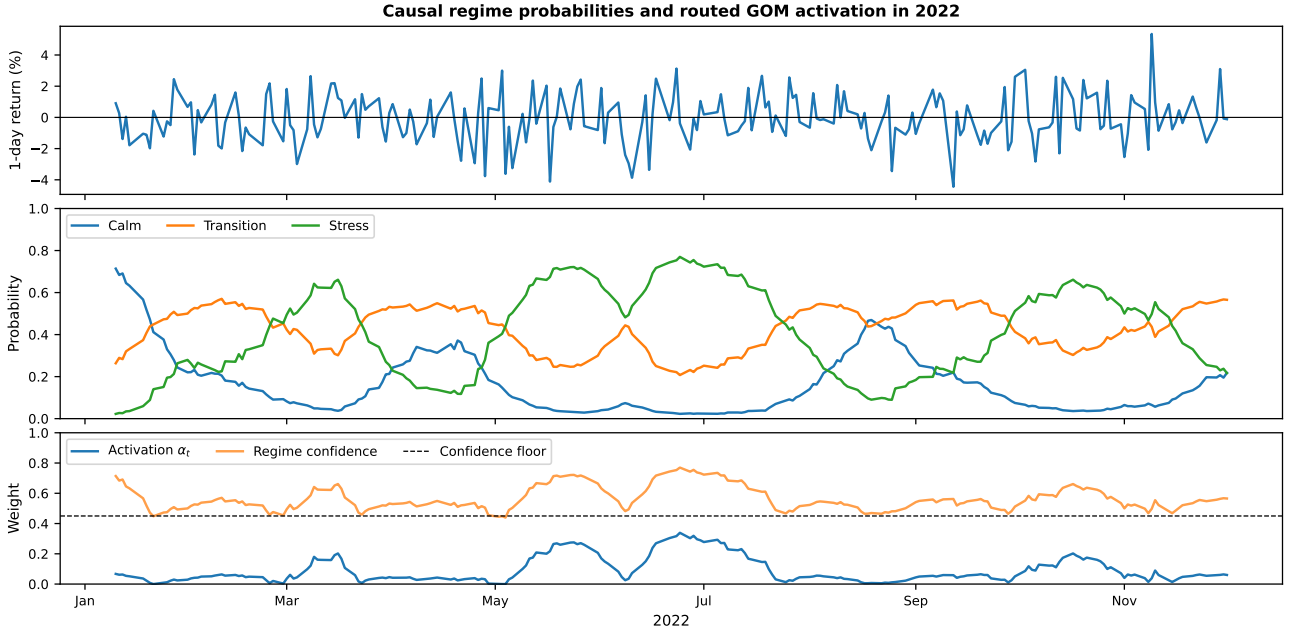


Figure 5: Regime Gating Dynamics. This visualization maps the causal regime probabilities alongside the routed GOM activation during the accepted 2022 experiment. The neural activation α_t scales upward exclusively when the system’s confidence in the current regime strongly exceeds the 0.45 floor and the specific probability-weighted residual cap remains positive.

Algorithm 1: Formulation of the Adaptive Statistical Model

Input: Training returns, dense quantile grid Q , forecast horizons $\mathcal{H}$, fixed EWMA baseline Q^A , adaptive models $\{Q^{(j)}\}_{j=1}^J$.

Output: Robust Statistical forecast Q^S alongside its fitted parameter state.

```

1 Partition training origins chronologically into distinct pool-fit, blend-selection, calibration, and confirmation blocks. // Ensures zero look-ahead bias during internal fitting
2 foreach  $h \in \mathcal{H}$  do
3   Estimate convex weights  $a_{h,j}$  strictly on the pool-fit block
4   Form combined adaptive pool:  $Q_{t,h}^P = \sum_j a_{h,j} Q_{t,h}^{(j)}$ 
5   Choose optimal blend parameter  $\beta_h$  on the blend-selection block
6   Estimate the spread scale  $s_h$  and shrunk quantile offsets  $\delta_h$  exclusively on the calibration block
7   Evaluate the resulting fixed candidate systematically on chronological confirmation episodes
8   Compute a one-sided circular-block-bootstrap lower bound. // Validates statistical significance against serial dependence
9   if confirmation performance, episode stability, and bootstrap bounds all clear strict thresholds then
10     $Q_{t,h}^S \leftarrow \mathcal{M}[m + s_h \{Q^A + \beta_h(Q^P - Q^A) - m\} + \delta_h]$  // Apply successful adaptive modifications
11  else
12     $Q_{t,h}^S \leftarrow Q_{t,h}^A$  exactly. // The robust EWMA anchor
13 return  $Q^S$  and  $(a, \beta, s, \delta)$ .
```

Statistically, the one-day Kupiec test failed to reject the null hypothesis at the 5% level; however, the five-day coverage and independence tests rejected the null hypothesis.

While the raw generative models successfully compressed these exception rates—particularly enhancing one-day forecasts—they failed to systematically resolve the clustering of five-day exceptions. Therefore, while their risk diagnostics confirm the structural efficacy of generative modeling, they remain statistically insufficient to justify retroactive promotion outside the established causal gating rules.

7 Discussion

The collective evidence from the development folds advocates for a nuanced, regime-dependent approach to neural enhancement. The 2022 forecasts produced by GOM and Tail-GAN validated the pre-test promotion rule, yielding improvements in mean pinball loss. Notably, the successful GOM correction

exerted its strongest influence during market transitions and stress states. Tail-GAN also achieved accepted status in 2024, though the realized improvement was smaller.

However, the final 2025 experiment profoundly shifts the paradigm of interpretation. While raw Tail-GAN and GOM ensembles objectively outperformed the EWMA baseline on average, the strict 2024 selector correctly identified their lack of transferability. This exposes a fundamental problem of mechanism selection. Applying a more permissive gating threshold would have certainly unlocked the impressive 2025 raw gains, but at the unacceptable cost of admitting highly unstable candidates during previous folds like 2020. Because the established thresholds were rigorously maintained without post-2025 observation tampering, this experiment provides a highly honest measurement of the risk-reward trade-off for machine learning in finance.

The Transformer architecture remained remarkably close to the EWMA baseline across all evaluated test periods. Its zero-initialized residual head and the strict benchmark guard

Algorithm 2: Neural Training and Probabilistic Promotion Gate

Input: Statistical forecast Q^S , historical panels X , target variables y , regime probabilities $\mathbf{p}$, random seed set $\mathcal{S} = \{11, 22, 33\}$.

Output: Final, gated, monotonically rearranged quantile surface Q^G .

```
1 foreach  $s \in \mathcal{S}$  do
2   Initialize and train an individual neural model leveraging only the isolated training block. // Train across different initializations
3   Export raw selection, tuning, confirmation, and out-of-sample test forecasts denoted  $\widehat{Q}^{(s)}$ 
4   On the isolated selection block, calculate non-negative seed weights  $\pi_s$  and regime scaling caps  $w_k$  subject to constraints  $\sum_s \pi_s = 1$  and  $\max_s \pi_s \leq 0.8$ 
5   Aggregate the neural outputs  $Q^M = \sum_s \pi_s \widehat{Q}^{(s)}$  and quantify the performance improvement relative to  $Q^S$ 
6   if the selection phase improvement fails to exceed the predetermined fixed threshold then
7     return  $Q^G \leftarrow Q^S$  exactly. // Immediately reject weak candidates before further testing
8   Hold parameters  $(\pi, \mathbf{w})$  rigorously fixed and evaluate across the tuning and confirmation blocks
9   Compute the effective sample of seeds  $N_{\text{eff}} = (\sum_s \pi_s^2)^{-1}$ , evaluate episode/regime improvements, verify VaR-calibration change, and compute the
    circular-block-bootstrap lower bound
10  if every single promotion criterion is successfully satisfied then
11    Calculate the causal activation variable:  $\alpha_t \leftarrow \text{clip}((c_t - c_0)/(1 - c_0), 0, 1) \sum_k p_{t,k} w_k$ 
12    Apply the bounded residual correction:  $Q^G \leftarrow M[Q^S + \alpha_t(Q^M - Q^S)]$ 
13  else
14     $Q^G \leftarrow Q^S$  exactly. // Reject due to instability, reverting to the baseline
15 return  $Q^G$ .
```

Algorithm 3: Chronological Development and Final Holdout Evaluation

Input: Development test folds $\{2020, 2022, 2024\}$ and the untouched final test fold 2025.

Output: Common-origin pinball scores, dependent uncertainty intervals, and promotion decisions.

```
1 foreach development test year do
2   Fit the statistical referencing solely the preceding training history
3   Train three unique seeds for each distinct neural family
4   Estimate the seed combinations and regime caps based purely on the preceding validation year
5   Apply the gating logic from Algorithm 2 unconditionally, blinding the system from test outcomes
6   Evaluate all final derived forecasts against the common test origins
7 Transition to the final experiment by retaining identical architectures, quantiles, seed sets, and predefined promotion thresholds
8 Execute fitting using data completely through 2023, reserving 2024 strictly for selection, tuning, and confirmation gating
9 Evaluate the resulting, pre-selected formal forecast a single time on the common 2025 test origins
10 Report the raw (ex post) candidate capabilities clearly separated from the formally gated (ex ante) final forecast
11 return Complete development and final-holdout empirical result tables.
```

effectively prevented catastrophic departures from the statistical forecast. Yet, despite comprehensive self-supervised learning, its generated representations failed to forge a persistent, gate-clearing advantage. Conversely, the scenario-based models displayed significantly larger raw gains alongside profound regime sensitivity, successfully improving raw VaR diagnostics in 2025.

Several limitations constrain these findings. First, extrapolating universal conclusions from four annual test periods fails to span the exhaustive spectrum of possible market regimes. Second, applying a standard square-root-of-time scaling adjustment to the reference distribution serves only as a mathematical approximation for deeply overlapping multi-day returns. Third, the regime probabilities in this framework operate as deterministic functions of a limited feature set, rather than dynamically evolving estimates derived from a latent-state transition model. Fourth, optimizing the Tail-GAN critic and GOM objectives over a finite strategy bank does not strictly guarantee equality across the complete multivariate path distribution. Finally, the considerable width of the 2025 raw confidence intervals reflects the constrained sample sizes, which could be considered as a limitation.

8 Conclusion

The robust probabilistic benchmark and the causal promotion rule helps generate a highly disciplined, scientifically sound framework for evaluating high-capacity financial forecasting

models. Over the development sample, GOM and Tail-GAN successfully improved the 2022 quantile forecasts, while Tail-GAN demonstrated a minor acceptable gain in 2024.

Future research would pivot toward refining the selection mechanism, such as developing broader historical confirmation episodes, integrating probabilistic latent-state models, and formulating bespoke objectives targeting multi-day exception dependence.

Data and Code Availability

The comprehensive repository accompanying this research contains full model implementations, configuration files, experiment runners, regression tests, graphical assets, and the original LaTeX source code. To respect proprietary licensing, raw market data and trained network checkpoints are excluded. However, the repository exhaustively documents the necessary source schema and execution commands required to reconstruct the processed panels using independently sourced data.

A Principal Implementation Settings

Table 5 records the principal dimensions, hyperparameters, and rigid promotion thresholds applied universally across all four reported test periods without alteration.

Table 1: Development-fold results evaluated on common origins. The Raw column demonstrates the ex post capability of the neural models, whereas the Final column displays the strictly gated forecast generated via pre-test selection.

Test	Model	EWMA	Raw	Final	Final gain	Promotion outcome
2020	Transformer	0.005182	0.005141	0.005182	0.000%	Rejected at selection stage
2020	Tail-GAN	0.005182	0.005070	0.005182	0.000%	Rejected by episode/regime confirmation
2020	GOM	0.005182	0.005074	0.005182	0.000%	Rejected: bootstrap lower bound non-positive
2022	Transformer	0.004126	0.004125	0.004126	0.000%	Rejected at selection stage
2022	Tail-GAN	0.004126	0.004123	0.004114	0.285%	Accepted
2022	GOM	0.004126	0.004086	0.004109	0.408%	Accepted
2024	Transformer	0.002045	0.002045	0.002045	0.000%	Rejected at selection stage
2024	Tail-GAN	0.002045	0.002065	0.002041	0.203%	Accepted
2024	GOM	0.002045	0.002065	0.002045	0.000%	Rejected at selection stage

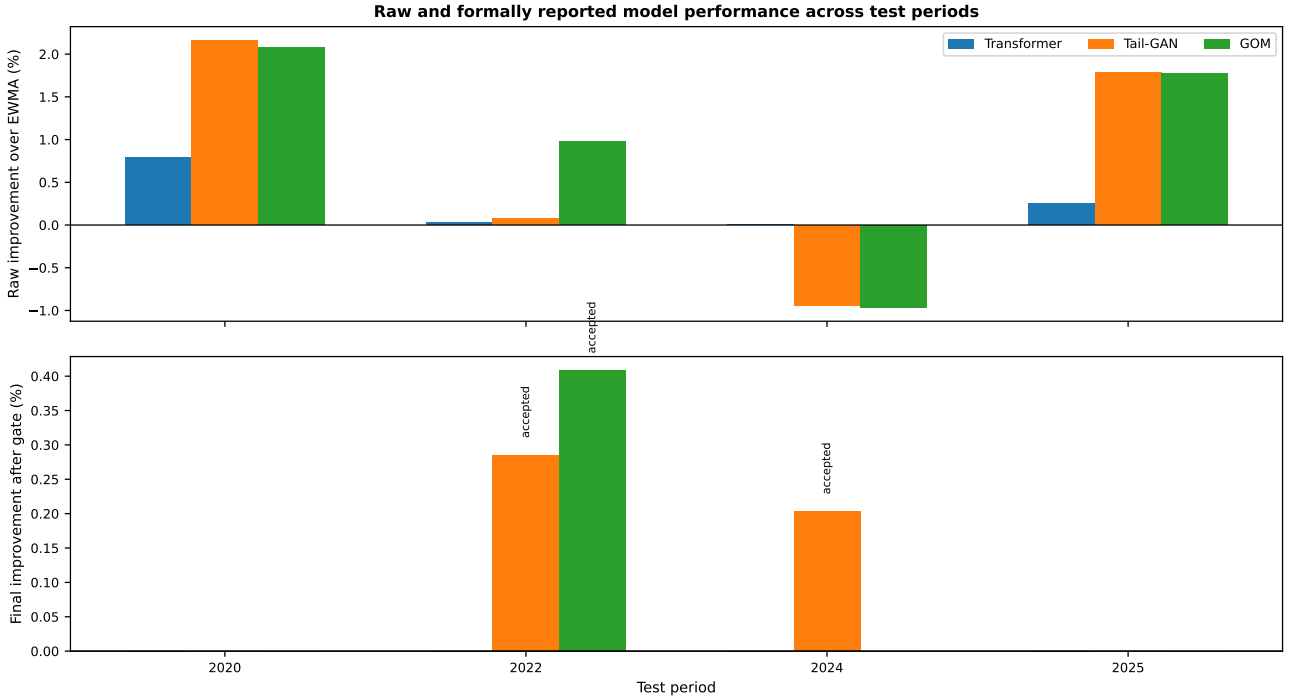


Figure 6: Cross-Period Model Viability. The upper panel highlights the diagnostic raw capability of the neural seeds, representing what the models learned. The lower panel displays the actual applied improvements after the strict pre-test gate governed activation.

B Interpretation of Raw and Formal Results

The raw ensemble denotes the equal- or validation-weighted aggregation of the three trained network seeds prior to any probabilistic routing intervention. It is reported to demonstrate the theoretical maximum of what the neural architecture managed to learn. Conversely, the formal forecast represents the rigid, real-world output mandated by the causal logic of Algorithm 2.

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Table 2: Untouched 2025 Test Results. Raw improvements describe the ex post realization of the models’ learning. Conversely, the formal forecast strictly reflects the deployment decision dictated by the pre-test 2024 gating mechanism.

Model	EWMA	Raw	Raw gain	Formal	Formal gain	Decision made purely from 2024 data
Transformer	0.0028594	0.0028522	0.252%	0.0028594	0.000%	Rejected: selection gain fell below initial threshold
Tail-GAN	0.0028594	0.0028082	1.790%	0.0028594	0.000%	Rejected: confirmation bootstrap lower bound < 0
GOM	0.0028594	0.0028088	1.770%	0.0028594	0.000%	Rejected: confirmation bootstrap lower bound < 0

Table 3: Pre-2025 Promotion Evidence. These metrics were estimated exclusively using 2024 data. Percentage values represent the relative improvement over the EWMA statistical model.

Model	Tuning	Confirmation	Positive episodes	Bootstrap lower bound	Promotion result
Transformer	0.000%	0.000%	—	0	Selection improvement below threshold
Tail-GAN	0.425%	−0.103%	1 of 4	-6.93×10^{-6}	Rejected
GOM	0.313%	−0.007%	2 of 4	-2.70×10^{-6}	Rejected

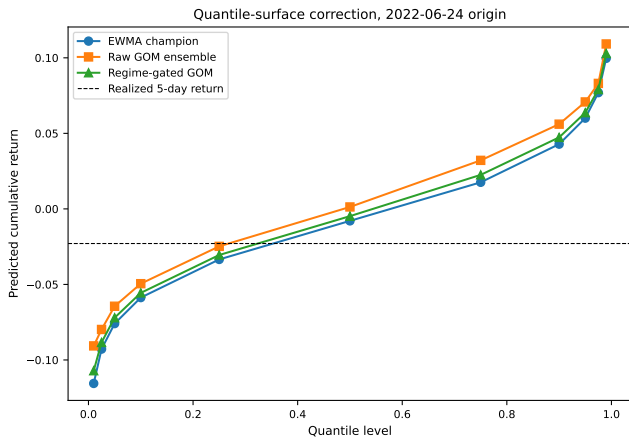
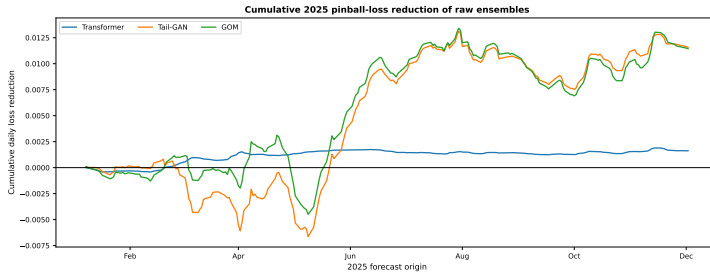
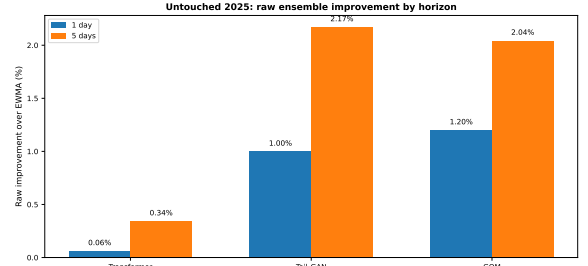


Figure 7: Quantile Surface Modification. A practical example of a 2022 five-day GOM quantile correction. The regime gate successfully improves the raw candidate model, while strictly maintaining the monotone quantile coherence.

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(a) Cumulative daily reduction tracking raw pinball loss.



(b) Raw improvement broken down by horizon length.

Figure 8: Raw 2025 Ensemble Diagnostics. The generative scenario models (Tail-GAN and GOM) demonstrated significantly stronger raw predictive improvements at the longer five-day horizon. However, these impressive ex post outcomes cannot alter the pre-2025 rejection issued by the causal gate.

Table 4: Untouched 2025 VaR Diagnostics. Exception rates evaluate the strict 5% lower-tail quantile boundary.

Forecast	Horizon	Pinball	Gain vs EWMA	VaR exceptions	Kupiec p	Independence p
EWMA formal	1	0.0018631	0.000%	7.52%	0.104	0.833
Transformer raw	1	0.0018619	0.062%	7.52%	0.104	0.833
Tail-GAN raw	1	0.0018445	0.999%	6.19%	0.426	0.885
GOM raw	1	0.0018407	1.201%	6.19%	0.426	0.827
EWMA formal	5	0.0038558	0.000%	10.18%	0.0016	1.3×10^{-14}
Transformer raw	5	0.0038425	0.343%	9.73%	0.0036	8.6×10^{-14}
Tail-GAN raw	5	0.0037720	2.172%	8.85%	0.016	2.0×10^{-10}
GOM raw	5	0.0037769	2.044%	8.41%	0.031	1.3×10^{-9}

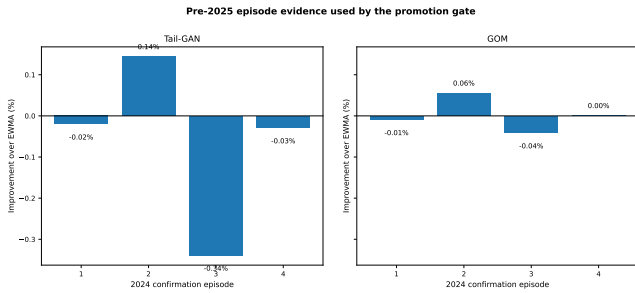


Figure 9: Episode-Level 2024 Validation Evidence. Evaluated prior to the 2025 test window, the conflicting positive and negative signs prevented both Tail-GAN and GOM from thematically satisfying the stringent promotion criteria.

Table 5: Fixed Architecture and Promotion Settings. Applied universally across all tested folds.

Component	Setting
History length	252 trading days
Value channels	95 plus explicit masks
Forecast horizons	1 and 5 days
Quantile levels	11, ranging from 0.01 to 0.99
Random seeds	11, 22, 33
Transformer dimensions	$d = 128$, 4 attention heads, 3 layers
Patch length / stride configurations	5/5, 21/10, 63/21
Self-supervised pretraining epochs	20
Supervised tuning epochs	up to 50
Scenario generation epochs	up to 65
Generator factor rank	4
Scenario draws per forecast origin	256
Bootstrap block length	10 contiguous origins
Bootstrap replications	2,000 iterations
Minimum allowable selection gain	0.15%
Minimum allowable confirmation gain	0.15%
Minimum required positive episodes	75%
Maximum allowable single seed weight	0.80
Minimum effective seeds metric	1.35
Regime activation confidence floor	0.45

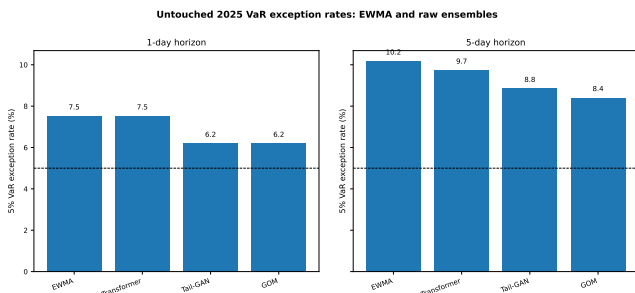


Figure 10: 2025 VaR Exception Rates. Observed violation rates at the 5% threshold against the nominal target (dashed line).